

MTH(622) Final term 2020

Q.1 The maximum value of the directional derivative of $\phi(x,y,z)$ is equal to the
 Magnitude of $\nabla \times \phi$ Magnitude of ϕ **Magnitude of $\nabla \phi$** None of these

Q.2 Which of the following property is not true? **true C) $\nabla \cdot \nabla \phi = \nabla^2 \phi$**

a) $\nabla \cdot (\vec{A} + \vec{B}) = \nabla \cdot \vec{A} + \nabla \cdot \vec{B}$ b) $\nabla \cdot (\phi \vec{A}) = \phi (\nabla \cdot \vec{A}) + \vec{A} \cdot (\nabla \phi)$

c) $\nabla \cdot \nabla \phi = \nabla^2 \phi$ d) $\nabla \cdot \vec{V} = \vec{V} \cdot \nabla$

Q.3 $\nabla \times \nabla \phi = ?$

$\nabla^2 \phi$ ϕ **0** None of these

Q.4 The stoke's theorem can be used to find which of the following?

- A) Area enclosed by a function in the given region
 B) Volume enclosed by a function in the given region
 C) Linear distance d) **Curl of the function**

Q.5 The functions in Green's theorem will be

Continuous derivatives discrete derivatives
Continuous partial derivatives discrete partial derivative

Q.6 If a time dependent vector function $\vec{A}$ is represented by $\vec{A}_f$ and $\vec{A}_r$ in fixed and rotating coordinate system, Then true $\left(\frac{d\vec{A}}{dt}\right)_f = \left(\frac{d\vec{A}}{dt}\right)_r + \omega \times \vec{A}_r$

$$\left(\frac{d\vec{A}}{dt}\right)_f = \left(\frac{d\vec{A}}{dt}\right)_r - \omega \times \vec{A}_r, \quad \left(\frac{d\vec{A}}{dt}\right)_f = \left(\frac{d\vec{A}}{dt}\right)_r + \omega \times \vec{A}_r, \quad \left(\frac{d\vec{A}}{dt}\right)_f = \vec{A}_r - \omega \times \vec{A}_r, \quad \left(\frac{d\vec{A}}{dt}\right)_f = \left(\frac{d\vec{A}}{dt}\right)_r = -\vec{A}_r \times \vec{A}_r$$

Q.7 The axes relative to which product of inertia are zero are called

Principal axes principal moment of inertia moment of inertia axes of rotation

Q.8 A 3×3 symmetrical matrix has real eigenvalues, which may be distinct or repeated.

Infinite Zero Two **Three**

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B) Volume enclosed by a function in the given region

C) Linear distance d) **Curl of the function**

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Continuous derivatives discrete derivatives

Continuous partial derivatives discrete partial derivative

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$$\left(\frac{d\vec{A}}{dt} \right)_f = \left(\frac{d\vec{A}}{dt} \right)_r - \omega \times \vec{A}_r, \left(\frac{d\vec{A}}{dt} \right)_f = \left(\frac{d\vec{A}}{dt} \right)_r + \omega \times \vec{A}_r, \left(\frac{d\vec{A}}{dt} \right)_f = \vec{A}_r - \omega \times \vec{A}_r, \left(\frac{d\vec{A}}{dt} \right)_f = \left(\frac{d\vec{A}}{dt} \right)_r = -\vec{A}_r \times \vec{A}_r$$

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Infinite

Zero

Two

Three

Q.9 The polar moment of inertia of circular section is about

X-X axis

Y-Y axis

Z-Z axis

Neutral axis

Q.10 The wheels of a moving car possess

Potential energy only

kinetic energy of translation only

Kinetic energy of rotation only

kinetic energy of translation and rotational both

Q.11 The moment of inertia of a circular section of base 'b' and height 'h' about an axis passing through its vertex and parallel to base is

$Bh^3/12$

$Bh^3/36$

$Bh^3/3$

$Bh^3/4$

Q.12 Moment of inertia of thin rod is given by formula of

$1/12 (ml^2)$

mr^3

$2 (mr)$

$2/5 (mr^2)$

Q.13 Cylinder has moment of inertia given by formula of

(ml^2)

mr

$3/2 (mr)$

$5 (mr^2)$

Q.14 Moment of inertia of sphere is given as

$2 (ml^2)$

r^2

$3/2 (r)$

$2/5 (mr^2)$

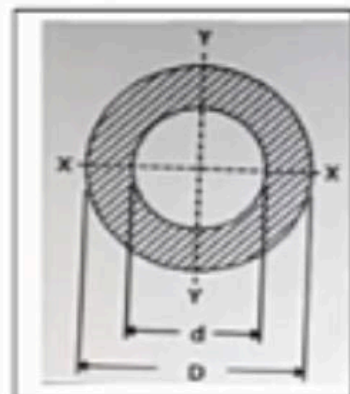
Q.15 Moment of inertia of a hollow circular section, as shown in the below figure about X-axis, is

$\pi/16 (D^2 - d^2)$

$\pi/16 (D^3 - d^3)$

$\pi/16 (D^4 - d^4)$

$\pi/64 (D^4 - d^4)$



Q.16 According to parallel axis theorem, the moment of inertia of a section about an axis parallel to the axis through center of gravity (i.e. IP) is given by _____.

(Where, A=area of the section, IG = Moment of inertia of the section about an axis passing through its C.G, and h = Distance between C.G and the parallel axis)

$IP = IG + Ah^2$

$IP = IG - Ah^2$

$IP = IG / Ah^2$

$IP = Ah^2 / IG$

Q.17 At pivot point, body with mass 'm' will have a torque by the

Force

circle

central point

radius of circle

Q.18 For a conservative holonomic dynamical system, the lagrangian L, kinetic energy and potential energy V are connected by.

$L = T + V$

$L = T - V$

$L = 2T + V$

$L = 2T - V$

Q.19 The periodic time of a particle with simple harmonic motion is _____ proportional to the angular velocity.

Directly

inversely

square root

none of these

Q.20 Work done is said to be zero, when

a) Some force acts on a body, but displacement is zero

b) No force acts on a body, but some displacement takes place

c) **Either a) or b)**

d) None of these

Q.21 The rate of change of displacement of a body is called?

Velocity

acceleration

Momentum

None of these

Q.22 For an irrotational flow

$\nabla \times \vec{v} = \vec{0}$

$\nabla \times \vec{v} \neq \vec{0}$

$\nabla \cdot \vec{v} = \vec{0}$

None of these

Q.23 The path of a projectile in a non-resistive medium is

Linear

circular

parabolic

hyperbolic

Directly

inversely

square root

none of these

Q.20 Work done is said to be zero, when

- a) Some force acts on a body, but displacement is zero
- b) No force acts on a body, but some displacement takes place
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Q.21 The rate of change of displacement of a body is called?

- Velocity
- acceleration**
- Momentum
- None of these

Q.22 For an irrotational flow

- $\nabla \times \vec{v} = 0$
- $\nabla \times \vec{v} \neq 0$
- $\nabla \cdot \vec{v} = 0$
- None of these

Q.23 The path of a projectile in a non-resistive medium is

- Linear**
- circular
- parabolic
- hyperbolic

Q.24 In a conservative field, the total energy of particle _____ throughout the motion?

- a) Varies rapidly
- b) does not remain constant
- c) remains constant
- d) none of these

Q.25 The path of projectile is called its

- Velocity
- acceleration
- trajectory**
- none of these

Q.26 The time taken by the particle to complete one oscillation is called?

- Amplitude
- time period
- frequency
- none of these

Q.27 Until a force acts on a body, its velocity is

- Zero
- constant
- increasing
- decreasing

Q.28 If there is no net force acting on body, then its acceleration is

- Zero
- constant
- increasing
- decreasing

Q.29 Total amount of mass and energy together in a system is

- Increasing
- decreasing
- Zero
- constant

Q.30 Measure of average kinetic energy of molecules is

- Temperature
- energy
- internal energy
- enthalpy

Q.33 If moment of inertia of a uniform circular cylinder of length b and radius a about x-axis through the center and perpendicular to the central axis is: $I_x = m(\frac{1}{4}a^2 + \frac{1}{12}b^2)$

Q.31 What is a degree of freedom of the system of rigid bodies?

Ans:



Q.32 To find the inertia matrix of uniform solid cuboid (parallelepiped) at one of its corner,

if $I_x = \frac{M}{3}(b^2 + c^2)$ then find I_y , and I_z .

Ans:

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Q.34 Write the expression for total energy of a particle executing Simple Harmonic motion.

Ans:



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Q.35 Two objects push off against one another starting from a stationary position. First object with mass 65 kg acquires a velocity of 2 m/s to the right. What velocity does the second object having mass 70 kg acquire?

Ans:

Q.36 Consider a small element of a uniform elliptic plate has moment of inertia $I = \delta m y^2$. Write down the expression for moment of inertia for the whole plate. Note do not solve the expression.

Ans:

Q.37 If a co-ordinate system rotates with angular velocity $\vec{\omega} = 2\hat{i} - 3\hat{j} + 5\hat{k}$ relative to the fixed co-ordinate system, and $\vec{A} = \sin t \hat{i} - \cos t \hat{j} + e^{-t} \hat{k}$ then find $\frac{d\vec{A}}{dt}$ with respect to the fixed point system. Where $\left(\frac{d\vec{A}}{dt}\right)_r = -\sin t \hat{i} + \cos t \hat{j} - 2e^{-2t} \hat{k}$ and $\vec{\omega} \times \vec{A} = (4e^{-2t} - 3\cos t)\hat{i} - (-2e^{-t} + 3\sin t)\hat{j} + (\cos t + 2\sin t)\hat{k}$

Ans:

Q.38 What is the impulse developed by a mass of 100 gm which changes its velocity from $i+j+k$ to $2i+2j+3k$ m/sec?

Ans:

Q.39 Find the moments of inertia of a ring of radius $2a$ about an axis through center.

Ans:

Q.40

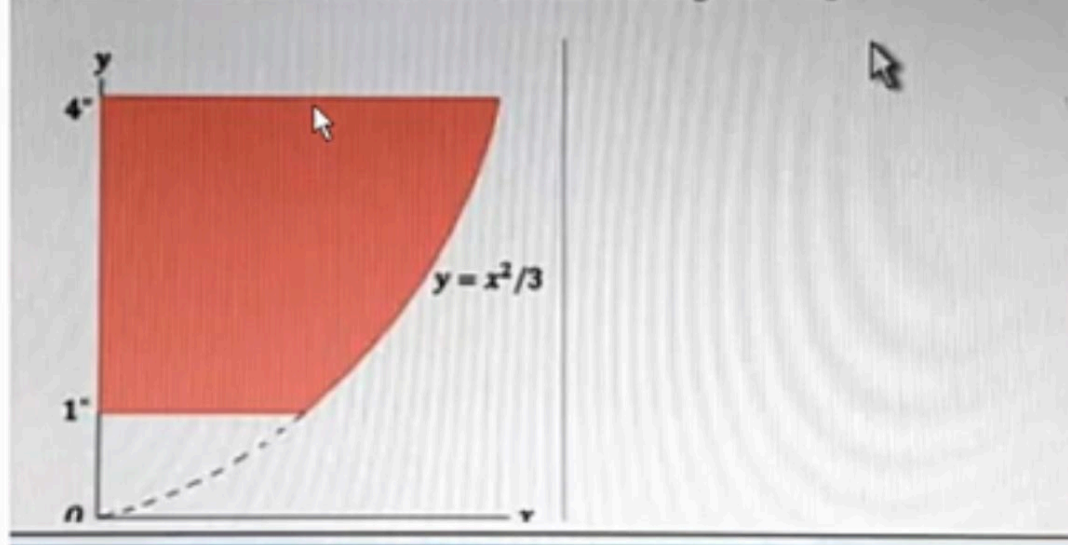
Calculate the moment of inertia of a uniform rod of length L about an axis perpendicular to the rod and passing through an end point. (Let the x-axis be chosen along the length of the rod, with origin at one end point let M and a be the mass and length of rod respectively and suppose the rod to be composed small elements.

Ans:

Q.41

Calculate the moment of inertia the shaded area given in the figure about y-axis

Calculate the moment of inertia of the shaded area given in figure about y-axis.



Whereas an elementary strip from the shaded area whose Moment of inertia is

Ans:

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